PHY202 Practice Exam on chapter(s): 22

1. $\varepsilon = 8.05 \times 10^{-4}\,\mathrm{V}\ \because \varepsilon = B\ell v$
2. $\dfrac{F}{\ell} = 476.\dfrac{\mathrm{N}}{\mathrm{m}}\ \because F = I\ell B\sin\theta, \theta = 90^{\circ}$
3. $B = 7.91 \times 10^{-5}\,\mathrm{T}\ \because r = \dfrac{mv}{qB}, m_e = 9.11 \times 10^{-31} kg, q_e = 1.60 \times 10^{-19} C$
4. $F = 32.1\,\mathrm{N}\ \because \dfrac{F}{\ell} = \dfrac{\mu_0 I_1 I_2}{2\pi r}$
5. $\tau = 1.11\,\mathrm{N}\cdot\mathrm{m}\ \because \tau = NIAB\sin\theta \therefore \tau_{max} = NIAB$
6. $B = 4.27 \times 10^{-5}\,\mathrm{T}\ \because r = \dfrac{mv}{qB}, m_e = 9.11 \times 10^{-31} kg, q_e = 1.60 \times 10^{-19} C$
7. $F = 7.62 \times 10^{-12}\,\mathrm{N}\ \because F = qvb\sin\theta, \theta = 90^{\circ}$
8. $\varepsilon = 1.72 \times 10^{-5}\,\mathrm{V}\ \because \varepsilon = B\ell v$
9. $F = 2.74 \times 10^{-7}\,\mathrm{N}\ \because F = qvb\sin\theta \therefore F_{max} = qvb$
10. $I = 48.7\,\mathrm{A}\ \because B = \dfrac{\mu_0 I}{2R}$
11. $I = 3.69\,\mathrm{A}\ \because B = \dfrac{\mu_0 I}{2\pi r}$
12. $\Delta y = 4.52 \times 10^{-12}\,\mathrm{m}\ \because F = qvb\sin\theta, F_{net} = ma, \vec{x}(t) = \vec{x}_0 + \vec{v}_0 t + \dfrac{1}{2}\vec{a}t^2$
13. $r = 0.944\,\mathrm{m}\ \because r = \dfrac{mv}{qB}, m_p = 1.67 \times 10^{-27} kg, q_p = 1.60 \times 10^{-19} C$